

FPGA Side-Channel Analysis

Midterm Presentation

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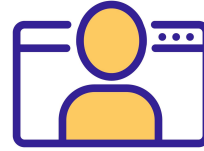
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Agenda

Project Summary

- Top Level Project Description/Problem Statement
- Project Objectives

Research Plan

- Research Plan Summary
- Deliverables
- Progress to Date

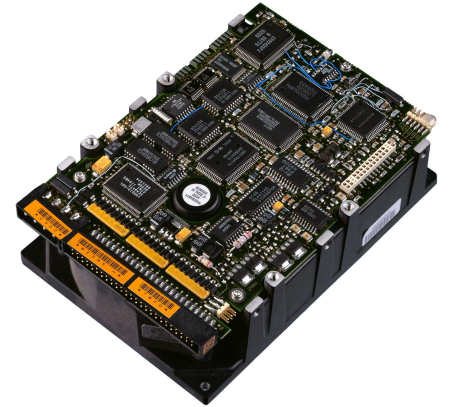
Summary of Results

Issues and Challenges

Project Summary

Top Level Project Description/Problem Statement

- FPGAs are fundamental hardware building blocks
 - Their security is crucial
- Goal: Provide a reliable means to detect malicious modifications to loaded programs



Project Objectives

- Provide a reliable means to detect malicious modifications made to an FPGA
- Load a benign program and a modified version of it onto FPGAs
- Compare data collected from each program



Research Plan

Research Plan Summary

- Learn HDL(s) and use Vivado to generate bitstreams
- Generate a benign and modified version of a hardware program
- Collect data through side-channel analysis
- Build a classifier with the data
- High- and low-synthesis versions of a bitstream will be created to test effects on side-channel analysis



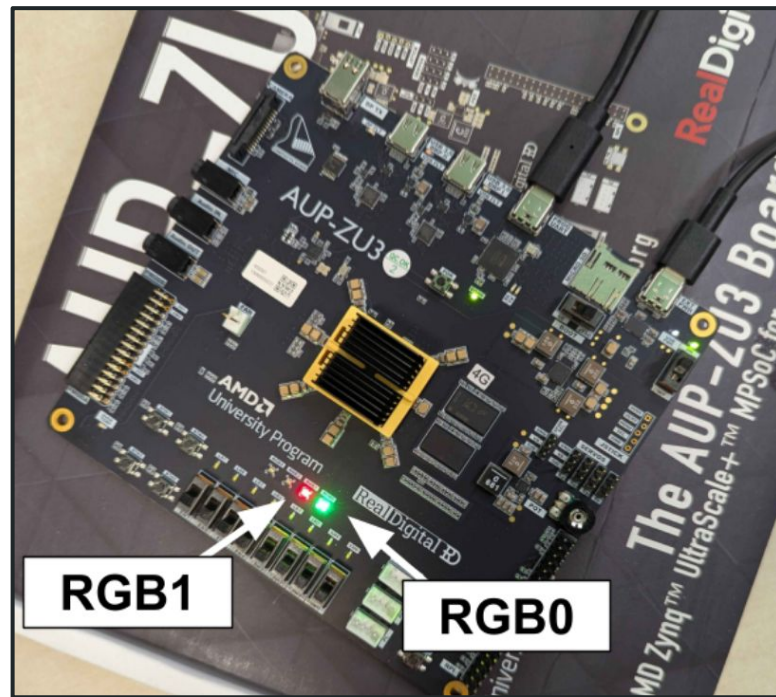
Deliverables

- Completed
 - 9/26 - Initial draft of benign traffic light program completed
 - 10/17 - Traffic light program completed and successfully loaded onto an FPGA
 - 10/24 - ChipWhisperer power analysis setup completed
 - 10/29 - Elevator program completed and successfully loaded onto an FPGA
 - 10/31 - Midterm report and presentation, HLS version of program completed and loaded
- In Progress
 - 11/7 - Completion and loading of a modified version of a benign program
 - 11/14 - Power consumption measurements for both the benign program and modified program
 - 11/14 - Electromagnetic emission measurements for both programs
 - 11/21 - Power and electromagnetic measurements for HLS program and low-level synthesis program
 - 11/28 - A report discussing the relationship between the two measurements, most likely within our final report



Progress to Date

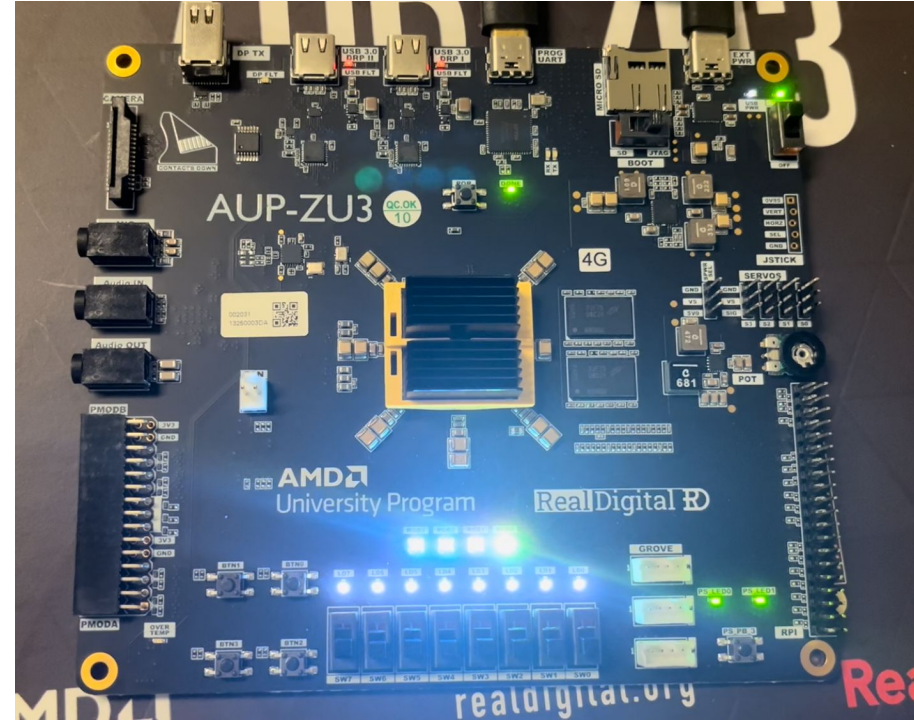
- Simple finite-state "traffic light" machine that performs the following:
 - RGB0, RGB1 both red for 3 seconds
 - RGB0 green for 20 seconds.
 - RGB0 yellow for 3 seconds.
 - RGB0, RGB1 both red for 3 seconds
 - RGB1 green for 20 seconds.
 - RGB1 yellow for 3 seconds.
 - (the cycle repeats.)



The FPGA we are using (RealDigital's AUP-ZU3 board), with the traffic light program running.

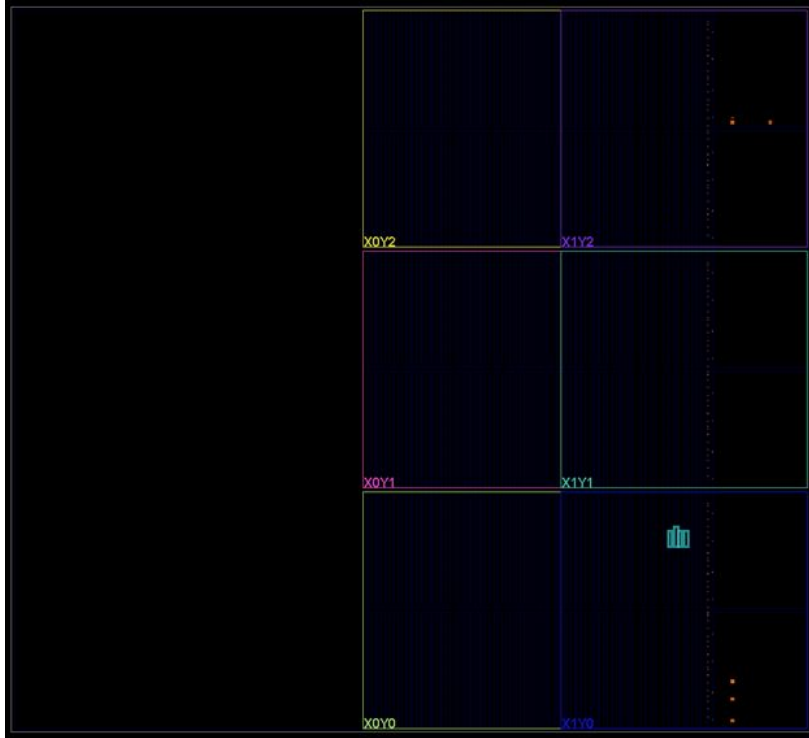
Progress to Date

- Low-level Verilog and HLS C++ versions of traffic light program
 - RGB0 turns green for 20 seconds
 - RGB0 turns yellow for 3 seconds
 - RGB0 turns red for 20 seconds
 - (the cycle repeats)
- Differences found in hardware implementation of low- and high-level versions

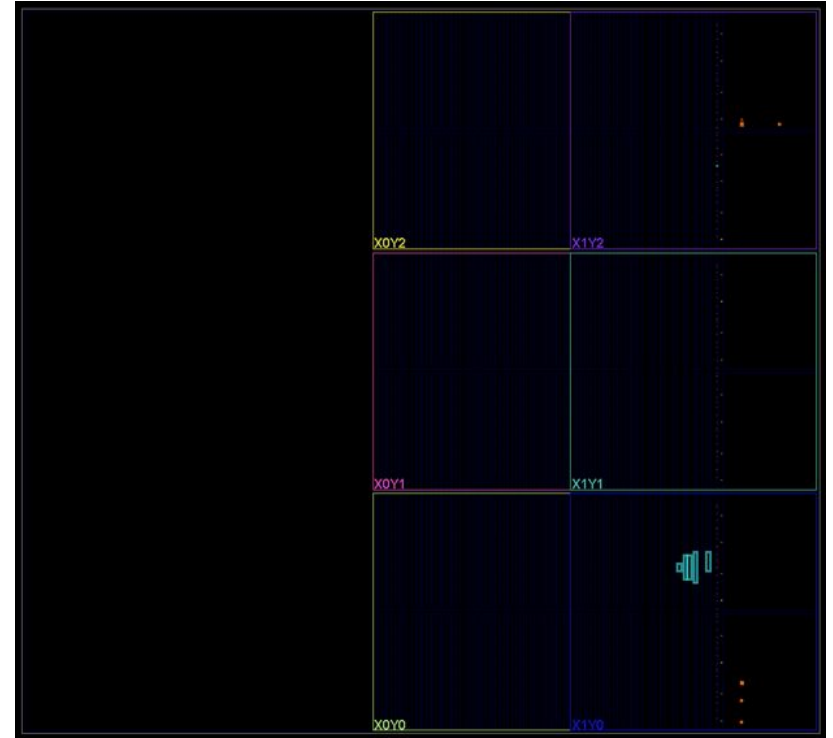


HLS traffic light implementation running on FPGA board in “yellow” state

Progress to Date



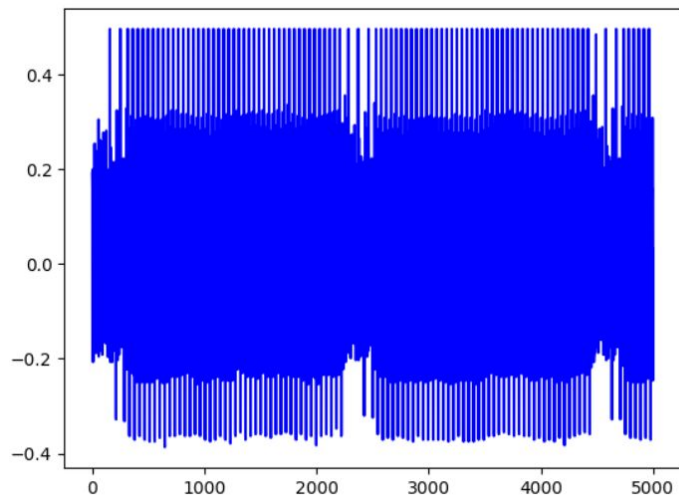
Hardware implementation with low-level Verilog synthesis



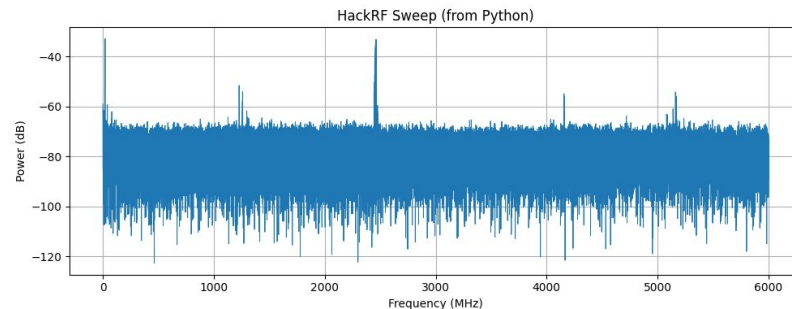
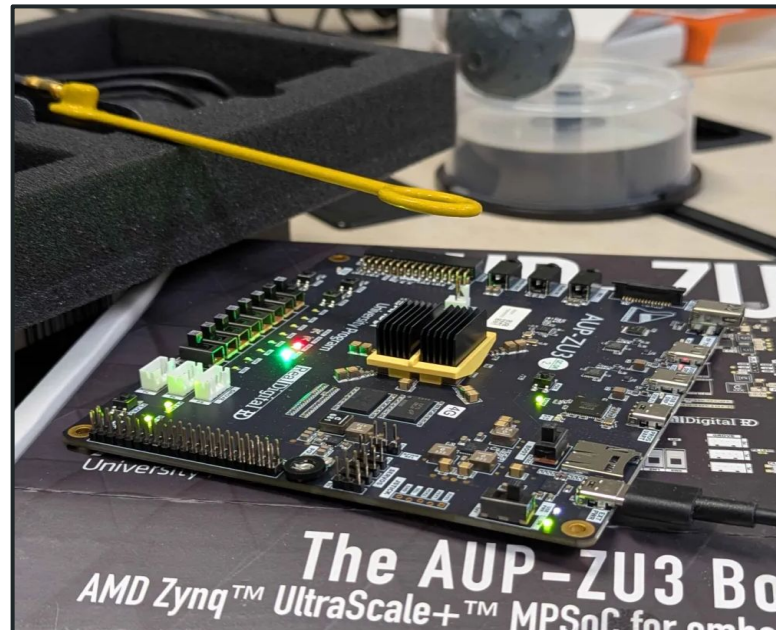
Hardware implementation with high-level C++ synthesis

Progress to Date

- Preliminary electromagnetic, power analysis setups
- Dummy ChipWhisperer power trace collection tests



Power trace collected in Jupyter Notepad with the ChipWhisperer Nano



Summary of Results

Summary of Results

- Developed and loaded several benign programs onto FPGA
- Discovered hardware implementation differences between low- and high-level synthesis
- Preliminary side-channel analysis setups (HackRF and ChipWhisperer)



Issues and Challenges

Issues and Challenges

- Temporary lack of hardware - FPGA boards
- Lack of prior experience in hardware design
- Hardware troubleshooting
- Anticipated Challenge: Hardware testbed setup



Appendix

Bibliography

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Acronyms

- **EM** - Electromagnetic
- **EM-SCA** - Electromagnetic side-channel analysis
- **FPGA** - Field-programmable gate array
- **FSM** - Finite state machine
- **HLS** - High level synthesis
- **SCA** - Side-channel Analysis
- **SDR** - Software-defined radio

Key Terms

- **Hardware Trojan:** A malicious modification to a circuit or logic that is disguised as being a part of the circuit, but has some harmful function.
- **FPGA:** AKA Field-Programmable Gate Array - is a configurable integrated circuit that can be continuously reprogrammed by users to perform some function.
- **Bitstream:** A file containing the data to be loaded onto the FPGA - this includes the main FPGA logic (program) and configuration data for routing, memory, and I/O.
- **Side-Channel Analysis:** Obtaining (potentially sensitive) information from a system through measuring the side effects of its main computations, such as power consumption and EM emissions.
- **Signal Processing:** Extracting information from a signal (sound, images, sensor data)
- **Supply Chain:** The life cycle of a hardware component, where at each point an attacker can add some malicious function

Q&A